

Quantum Mechanics III

Lecture 12

Control and Error Channels

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$U = Texp[-i \int H(t)dt/\hbar]$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$\overline{H} = \Omega_1 + \Omega_2 + \dots$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$P(m) = Ap^m + B$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 12; QDC control block; Instrument Regression guide